# EMCH 260
# SOLID MECHANICS

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# Stress Transformation

# 9

## Chapter 9.1

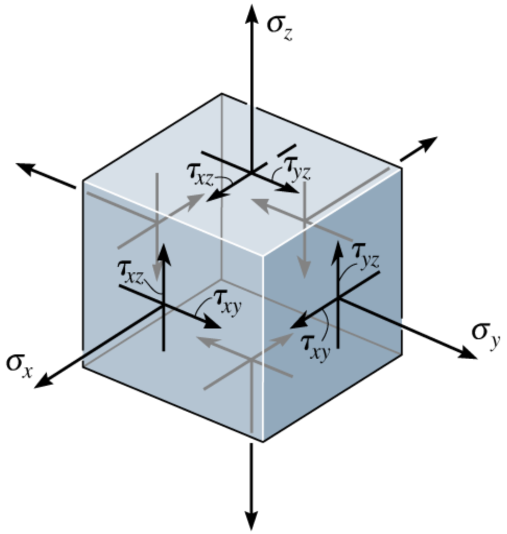
- ✓ Understand the plane stress state and know how to use an element to represent stress state at a point
- ✓ Know how to do plane-stress transformation
- ✓ Understand and use general equations of plane-stress transformation
- ✓ Understand principal stresses
- ✓ Know how to calculate principal stresses
- ✓ Know how to calculate maximum in-plane shear stress

# APPLICATIONS OF COMBINED LOADINGS

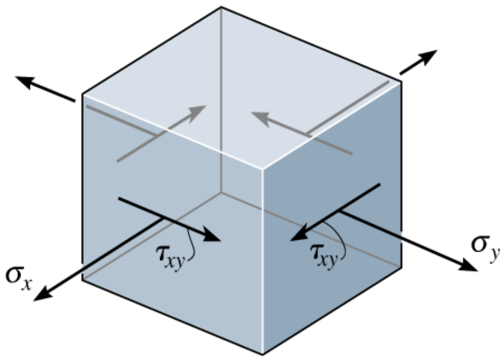
The cracks in this concrete beam were caused by tension stress, even though the beam was subjected to both an internal moment and shear. The stress transformation equations can be used to predict the direction of the cracks, and the principal normal stresses that caused them.

# GENERAL STATE OF STRESS AT A POINT

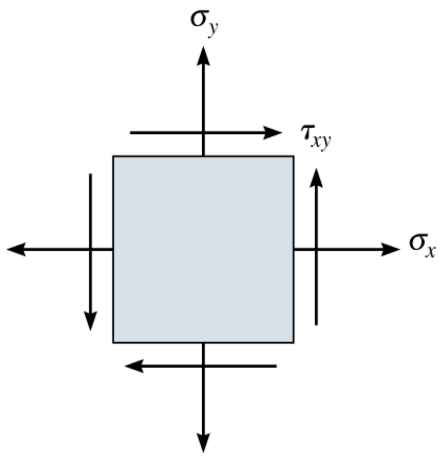
**General state of stress at a point is characterized by six independent normal and shear stress components; $\sigma_x$, $\sigma_y$, $\sigma_z$, $\tau_{xy}$, $\tau_{yz}$, and $\tau_{zx}$**




**General State of stress**



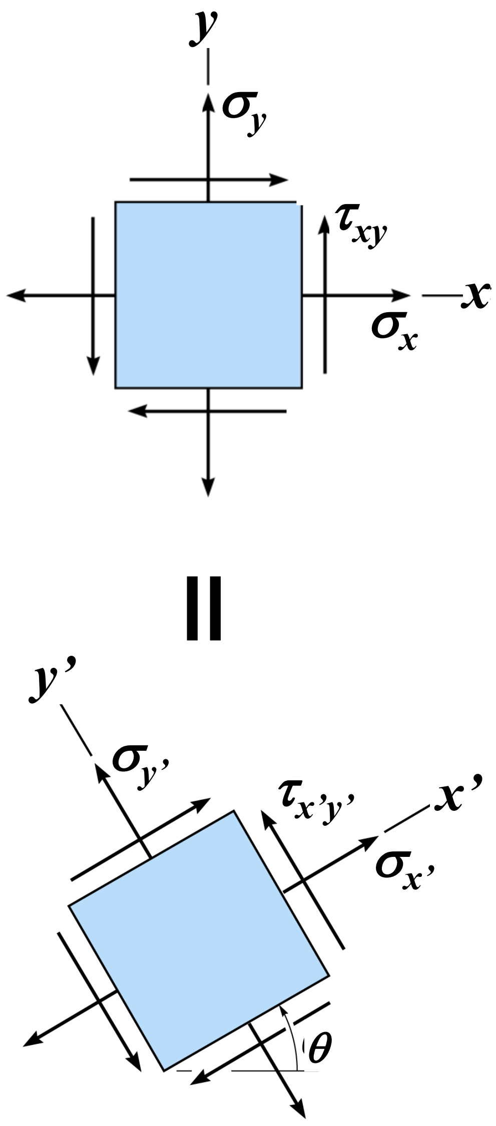

**Plane stress (a simplification)**




**Plane stress (two dimensional view)**

***Refer to definition of "stress" in lecture notes Ch1.2 p.21-24***

# ORIENTATION DEPENDENT STRESS COMPONENT




**If we rotate the element of the point in different orientation, we will have different values of the stresses**

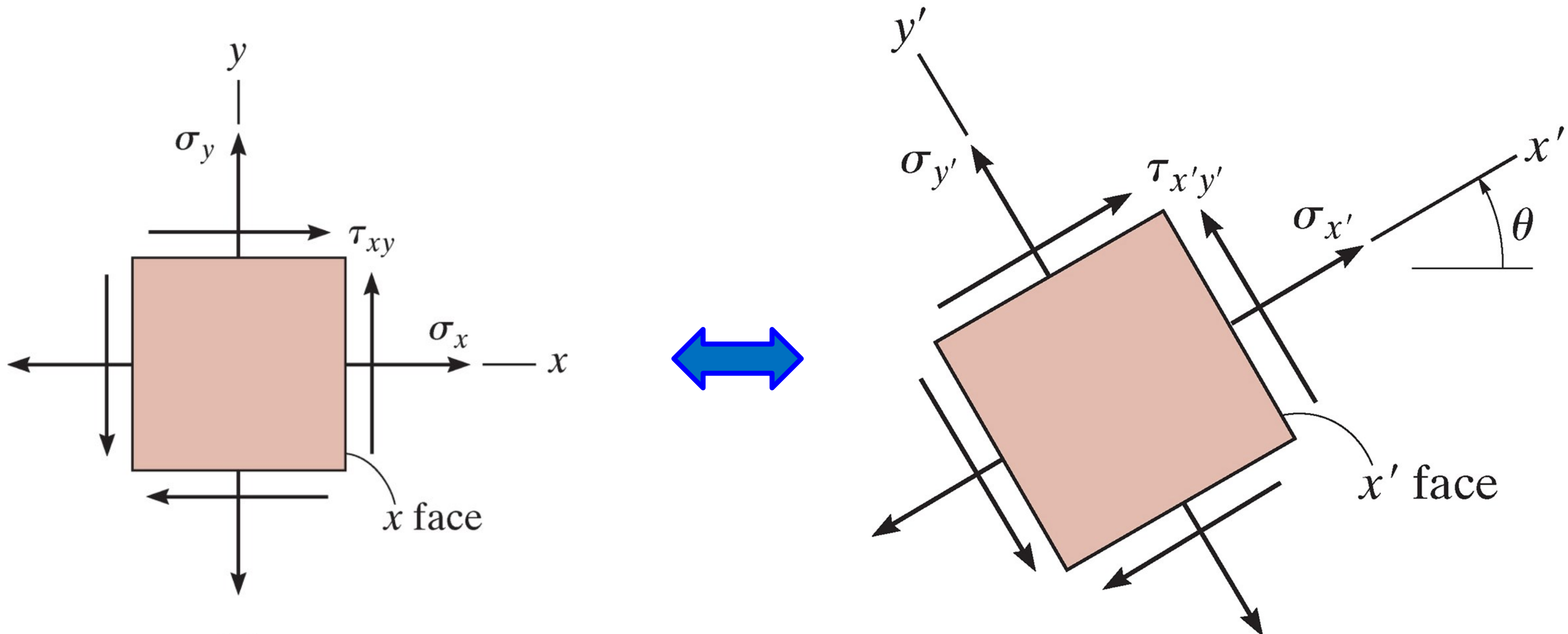
**The stresses are now $\sigma_{x'}$, $\sigma_{y'}$ and $\tau_{x'y'}$**

**The stress components can be transformed from one orientation of an element to the element having a different orientation**

# PROCEDURE FOR ANALYSIS

## <u>Step 1:</u> New Coordinate System

- Establish new coordinate system *Ox'y'* and draw the corresponding stress component

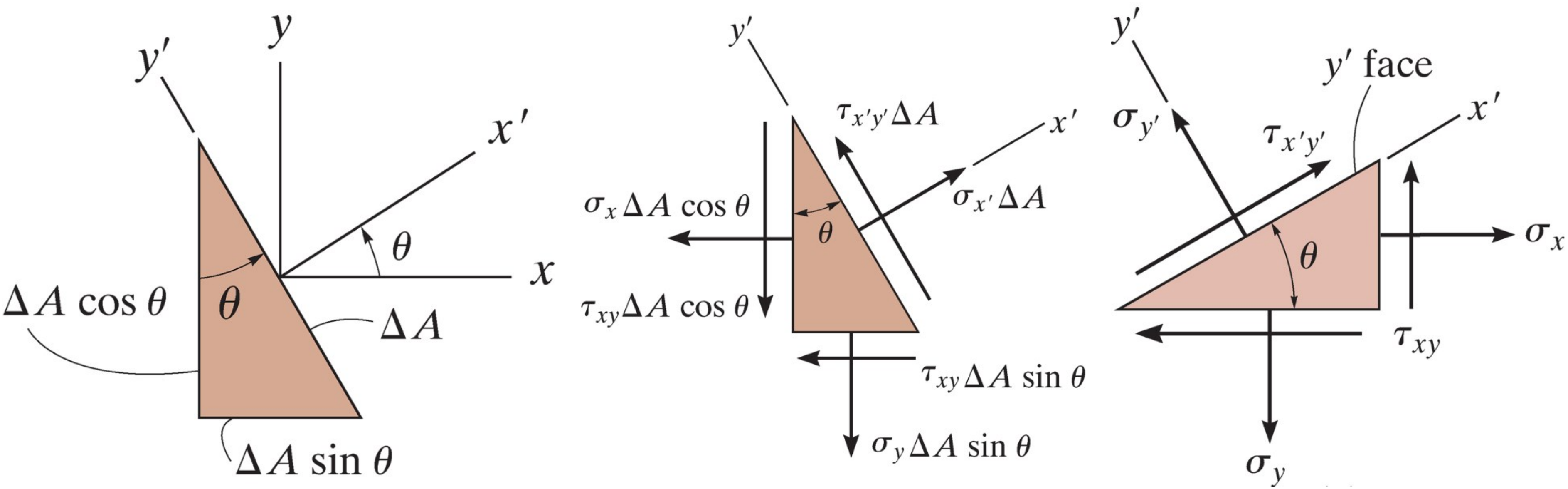



(a) (b)

# PROCEDURE FOR ANALYSIS, CONT'D

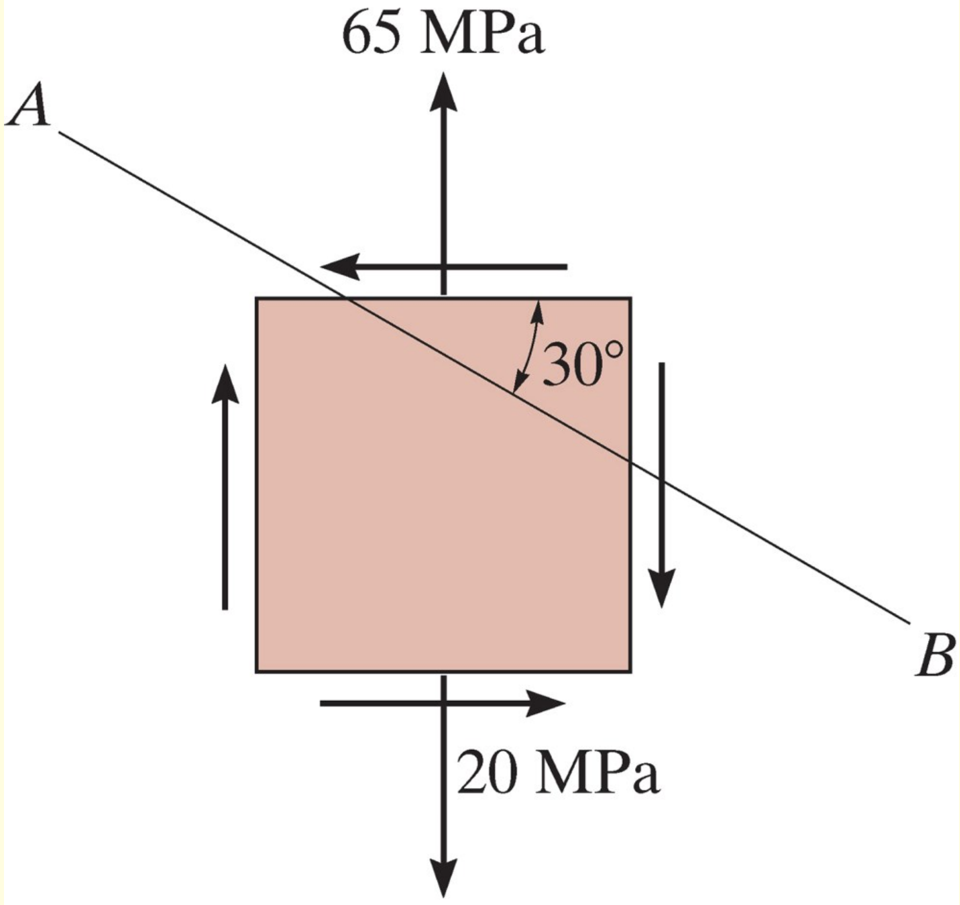
## Step 2: Free-Body Diagram

- Draw the free-body diagram of the segment by multiplying the stress components on each face by the area upon which they act
- Apply force equation of equilibrium will yield the new stress component



# IN-CLASS EXAMPLE 1

The plane-stress state of a point is represented by the element with normal stress and shear stress shown. Determine the stress components acting on the inclined plane *AB*.

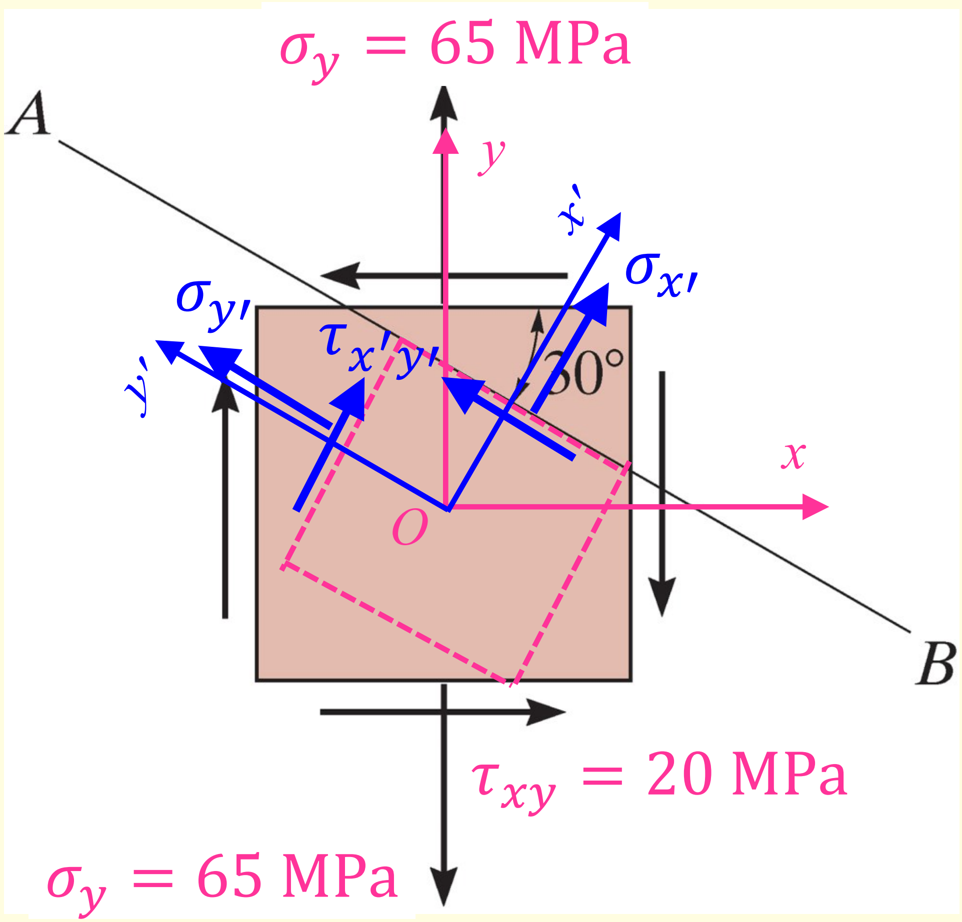


# IN-CLASS EXAMPLE 1, Cont'd

## Solutions

### Step 1: New Coordinate System

Establish new coordinate system *Ox'y'* and draw the corresponding stress component



# IN-CLASS EXAMPLE 1, Cont'd

## Solutions, Cont'd

### Step 2: Free-Body Diagram

- Apply force equation of equilibrium will yield the new stress component

Area that stress acts on

Force projected to *coordinate axis*

$$\sum F_{x'} = 0$$

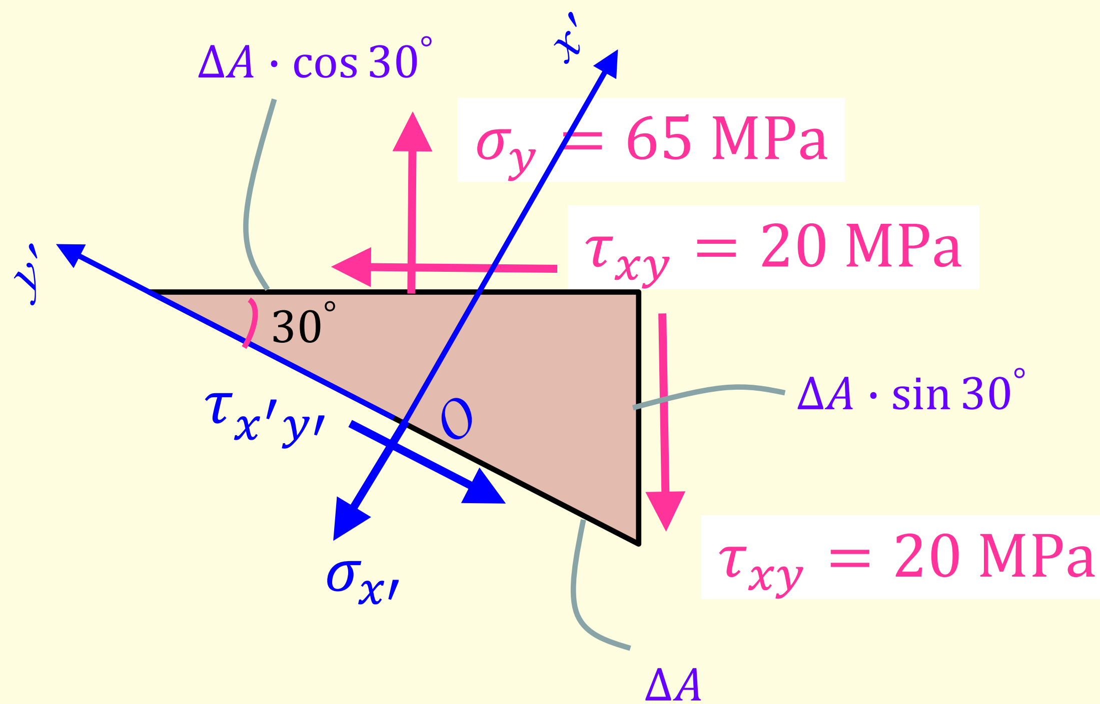
$$\rightarrow \left(\sigma_y \boxed{\cdot \Delta A \cdot \cos 30^\circ}\right) \boxed{\cdot \cos 30^\circ} - \left(\tau_{xy} \cdot \Delta A \cdot \cos 30^\circ\right) \cdot \sin 30^\circ - \left(\tau_{xy} \cdot \Delta A \cdot \sin 30^\circ\right) \cdot \cos 30^\circ - \sigma_{x'} \cdot \Delta A = 0$$

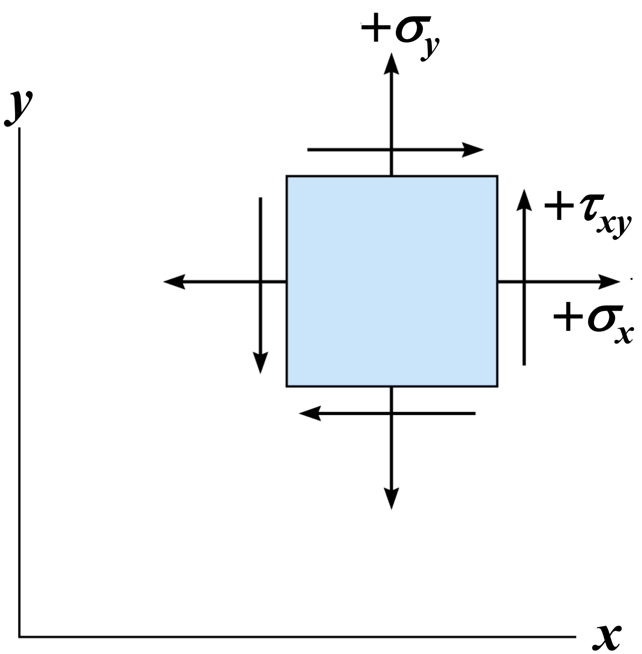
$$\rightarrow \sigma_{x'} = 31.43 \text{ MPa} \quad \textbf{(Ans.)}$$

$$\sum F_{y'} = 0$$

$$\rightarrow \left(\sigma_y \cdot \Delta A \cdot \cos 30^\circ\right) \cdot \sin 30^\circ + \left(\tau_{xy} \cdot \Delta A \cdot \cos 30^\circ\right) \cdot \cos 30^\circ - \left(\tau_{xy} \cdot \Delta A \cdot \sin 30^\circ\right) \cdot \sin 30^\circ - \tau_{x'y'} \cdot \Delta A = 0$$

$$\rightarrow \tau_{x'y'} = 38.15 \text{ MPa} \quad \textbf{(Ans.)}$$



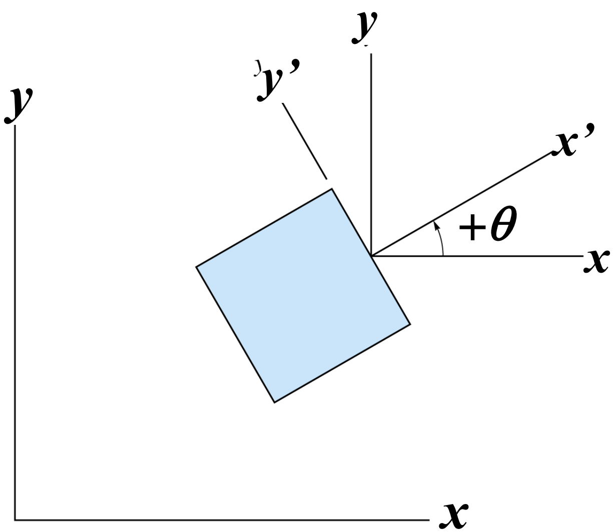



## Sign Convention

**A normal or shear stress component is *positive* provided it acts in the *positive* coordinate direction on the *positive* face of the element,**

**or it acts in the *negative* coordinate direction on the *negative* face of the element**

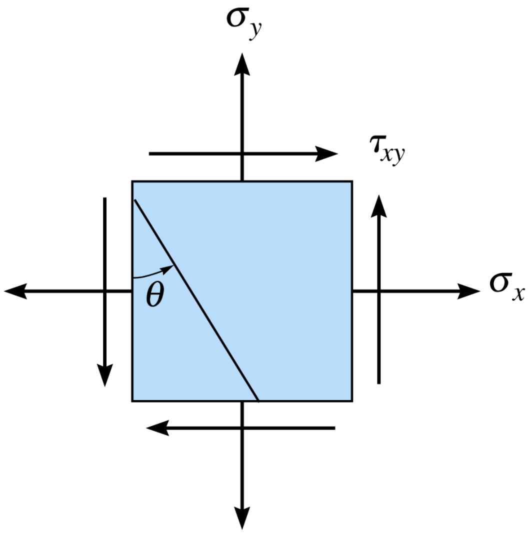
# ORIENTATION OF THE INCLINED PLANE



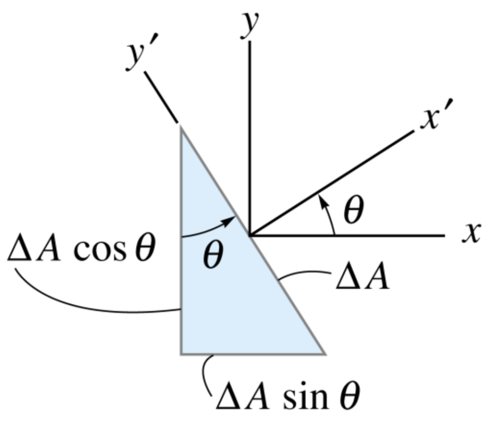

**The orientation of the inclined plane, on which the normal and shear stress components are to be determined, will be defined using the angle $\theta$**

**The angle $\theta$ is measured from the positive $x$ to the positive $x'$, i.e. The angle $\theta$ is *positive counterclockwise*.**

# GENERAL EQUATION OF PLANE–STRESS TRANSFORMATION




(a)

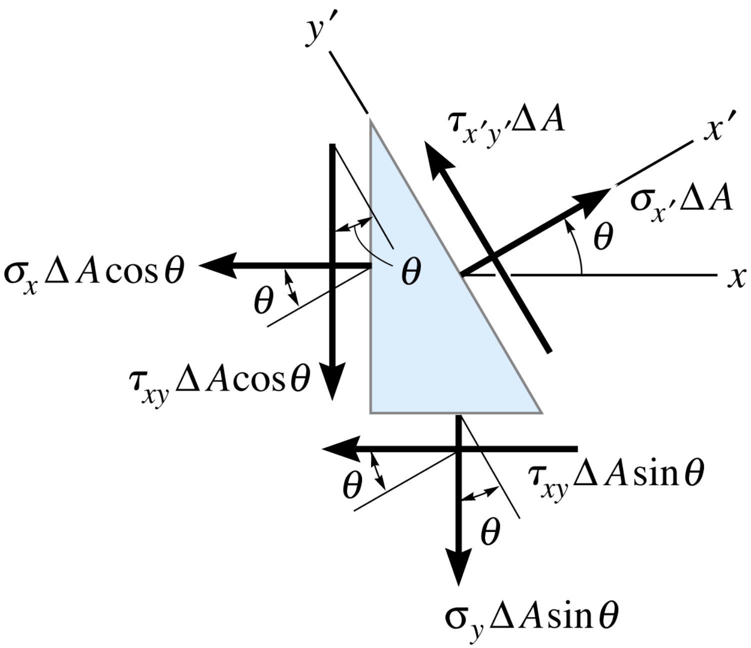



(b)

**The element in Fig.(a) is sectioned along the inclined plane and the segment shown in Fig.(b) is isolated.**

**Assuming the sectioned area is $\Delta A$, then the horizontal and vertical faces of the segment have an area of $\Delta A \cdot \sin\theta$ and $\Delta A \cdot \cos\theta$, respectively**

# GENERAL EQUATION OF PLANE–STRESS TRANSFORMATION, CONT'D




## Resulting *free-body diagram*

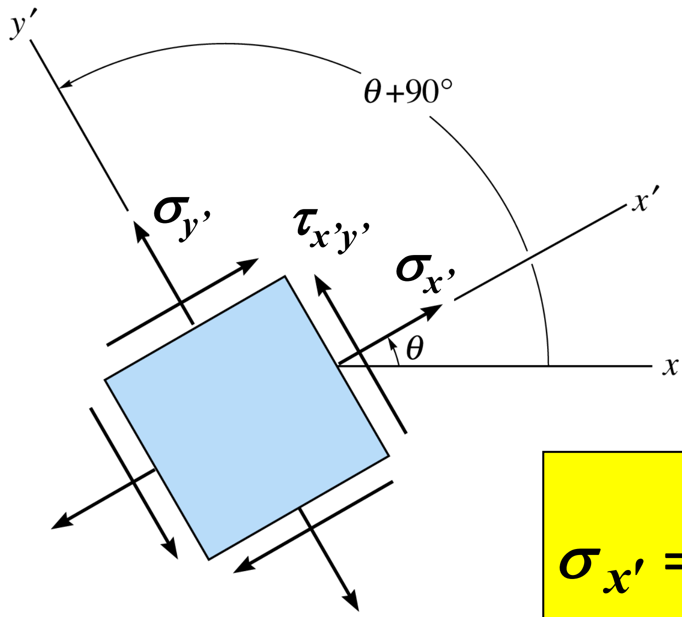
**The unknown normal and shear stress components in the inclined plane, $\sigma_{x'}$ and $\tau_{x'y'}$, can be determined from the equations of force equilibrium**

$$\sum F_{x'} = 0 \qquad \sum F_{y'} = 0$$

**We get**

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

# GENERAL EQUATION OF PLANE–STRESS TRANSFORMATION, CONT'D




**Three stress components, $\sigma_{x'}$, $\sigma_{y'}$ and $\tau_{x'y'}$, oriented along the $x'$, $y'$ axes**

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2}\cos 2\theta - \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

# REVISIT IN-CLASS EXAMPLE

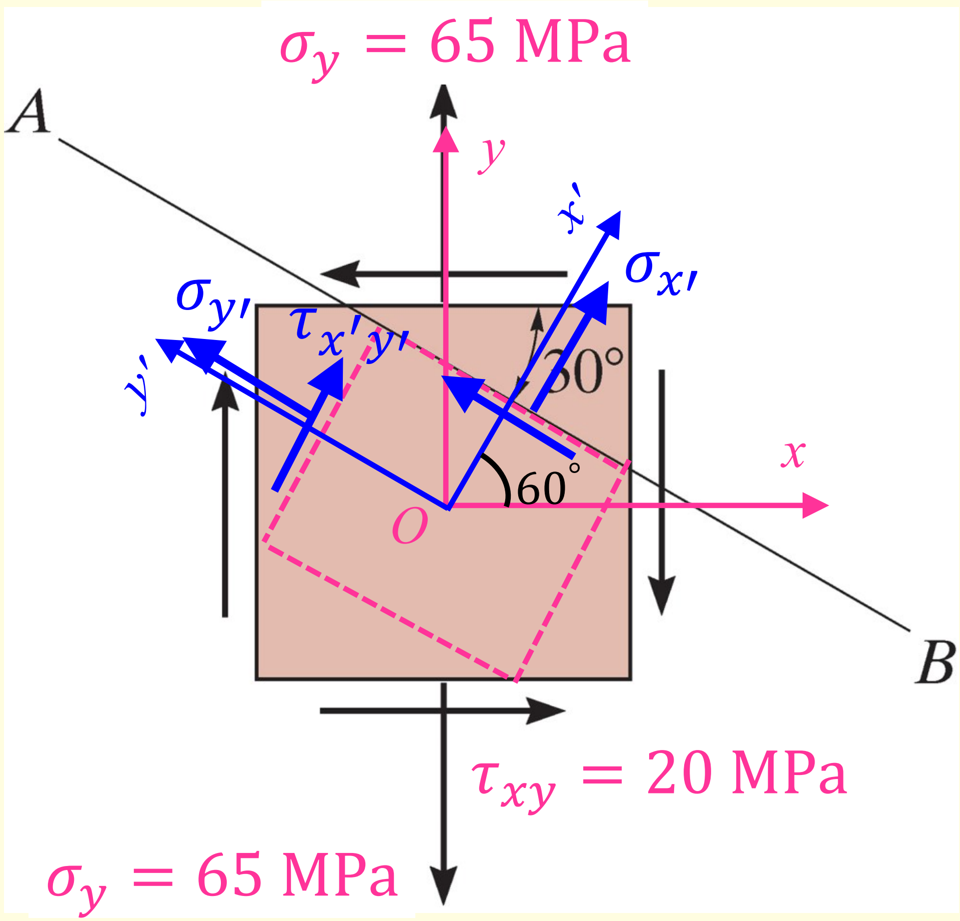
## Solutions

Revisit the aforementioned example:

$\boldsymbol{\sigma_x = 0}$
$\boldsymbol{\sigma_y = 65\ Mpa}$
$\boldsymbol{\tau_{xy} = -20\ Mpa}$
$\boldsymbol{\theta = 60^{\circ}}$

Using the general equations of plane-stress transformation ($\theta = 60^{\circ}$), we obtain:

$\boldsymbol{\sigma_{x'} = +31.43\ \mathrm{MPa}}$
$\boldsymbol{\sigma_{y'} = +33.57\ Mpa}$
$\boldsymbol{\tau_{x'y'} = +38.15\ \mathrm{MPa}}$




The answers are exactly the same as before!

The positive sign means the presumed directions are correct.

# IN-PLANE PRINCIPAL STRESSES

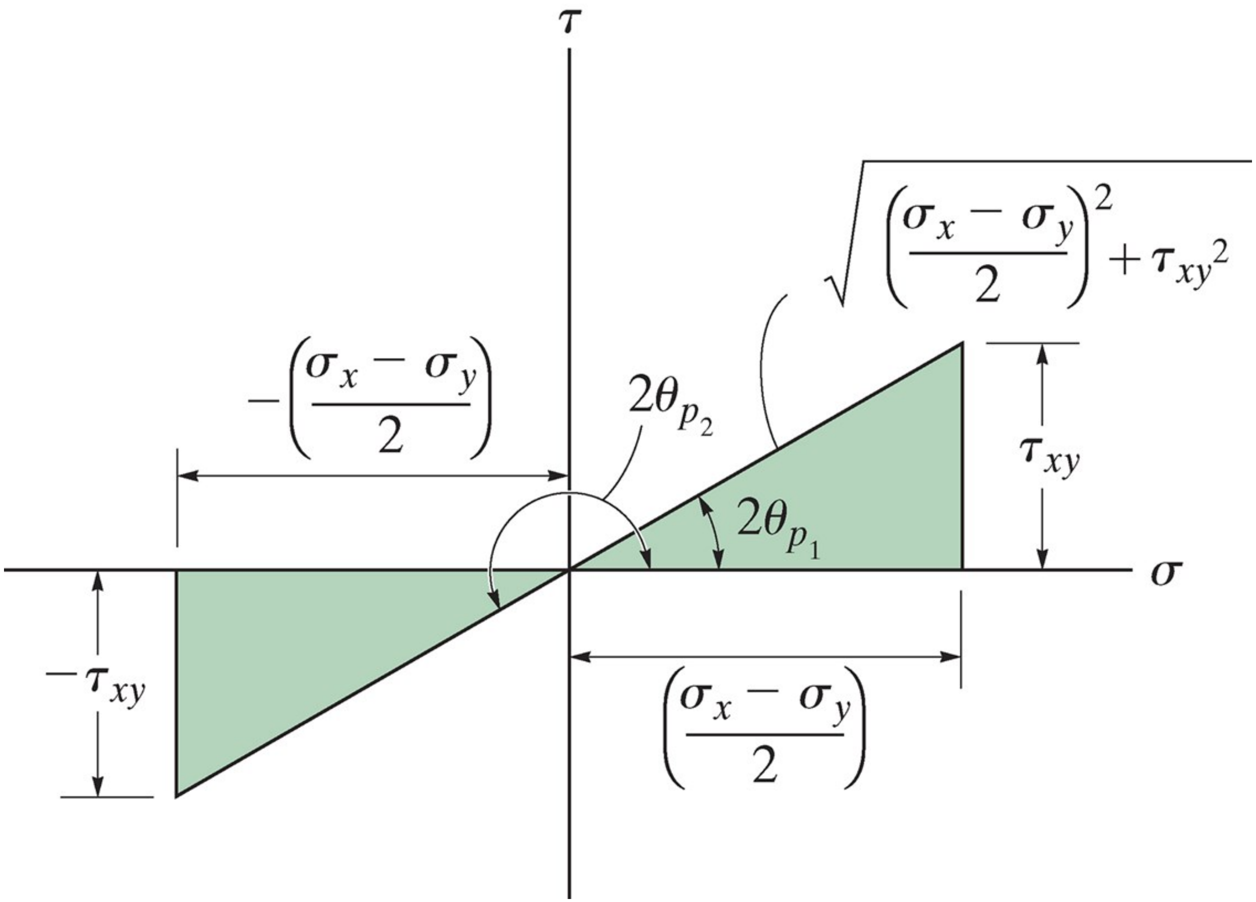
**To determine the maximum and minimum normal stress, we must differentiate equation of $\sigma_{x'}$ with respect to $\theta$ and set the result equal to zero. This gives**

$$\frac{d\sigma_{x'}}{d\theta} = -\frac{\sigma_x - \sigma_y}{2}(2\sin 2\theta) + 2\tau_{xy}\cos 2\theta = 0$$

**Solving this equation, we obtain the orientation $\theta = \theta_p$ of the planes of maximum and minimum normal stress**

$$\boxed{\tan 2\theta_p = \frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2}}$$

# IN-PLANE PRINCIPAL STRESSES, CONT'D




**The solution has two roots: $\theta_{p1}$ and $\theta_{p2}$**

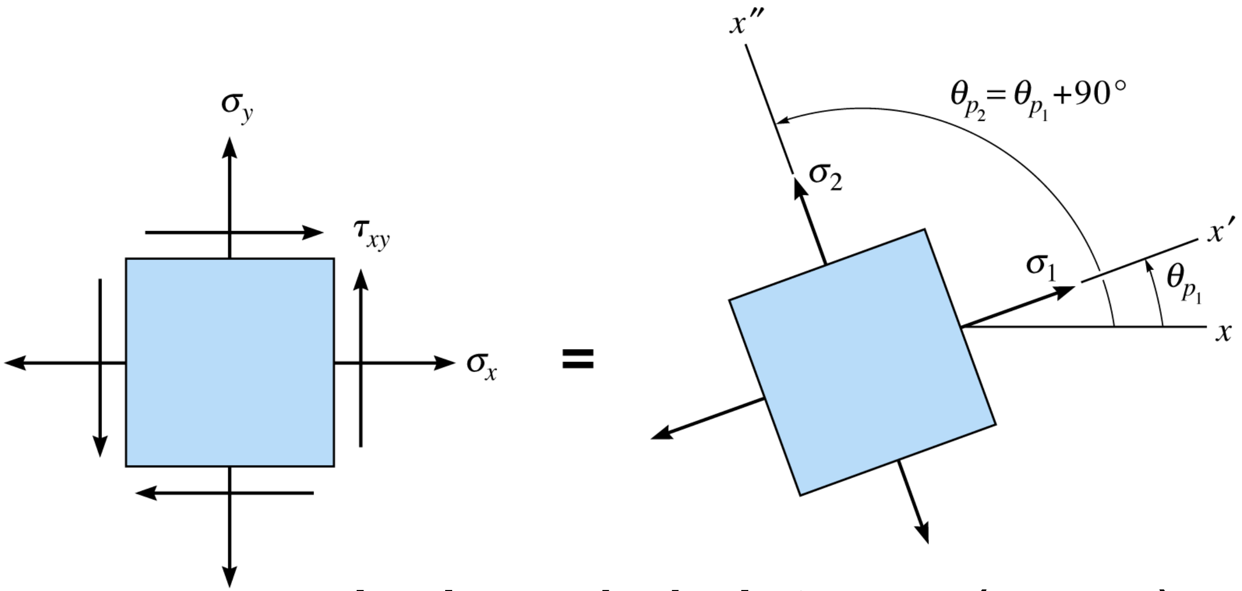
$$2\theta_{p2} = 2\theta_{p1} + 180^{\circ}$$

**or**

$$\theta_{p2} = \theta_{p1} + 90^{\circ}$$

**Based on the equation of tan2$\theta_p$ above, we can construct two shaded triangles as shown in the figure**

# IN-PLANE PRINCIPAL STRESSES, CONT'D




**In-plane principal stresses ($\sigma_1 \geq \sigma_2$)**

## Principal Planes

$$\tau_{x'y'} \equiv 0$$

***No shear stress acts on the principal planes***

***In other words, if there is no shear stress component on the element, you have already found the principle planes!***

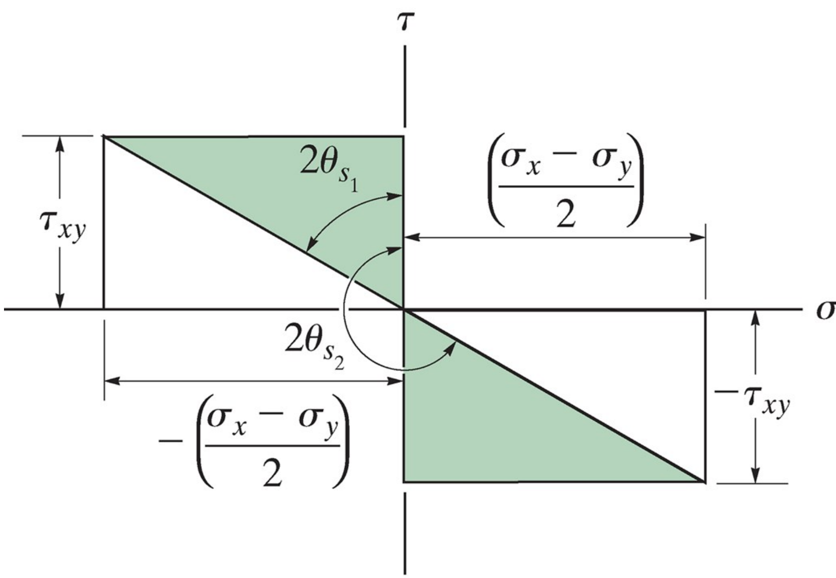
**Principal stresses = Maximum and minimum normal stress**

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

# MAXIMUM IN-PLANE SHEAR STRESS

**Similarly, to get the maximum shear stress, we must differentiate equation of $\tau_{x'y'}$ with respect to $\theta$ and set the result equal to zero. This gives**

$$\tan 2\theta_s = \frac{-\left(\sigma_x - \sigma_y\right)/2}{\tau_{xy}}$$




**The solution has two roots; $\theta_{s1}$ and $\theta_{s2}$**

$$2\theta_{s2} = 2\theta_{s1} + 180^\circ$$

**or**

$$\theta_{s2} = \theta_{s1} + 90^\circ$$

$$\tan 2\theta_p = -1/\tan 2\theta_s$$

$$\left|2\theta_p - 2\theta_s\right| = 90^\circ \quad \text{or} \quad \left|\theta_p - \theta_s\right| = 45^\circ$$

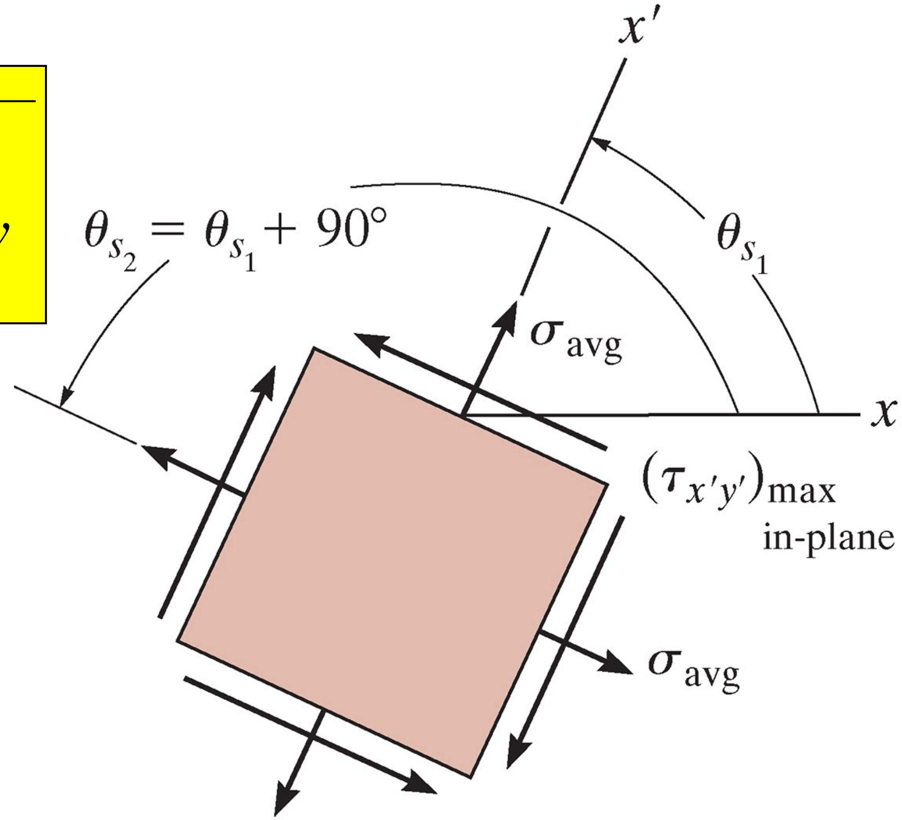
# MAXIMUM IN-PLANE SHEAR STRESS, CONT'D

**The maximum shear stress can be found by taking the trigonometric values of sin $2\theta_s$ and cos $2\theta_s$ from the figure and substituting them into equation of $\tau_{x'y'}$. The result is**

$$\tau^{max}_{in-plane} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

**Substituting the values for sin $2\theta_s$ and cos $2\theta_s$ into equation of $\sigma_{x'}$, we see that *there is also a normal stress on the planes of maximum in-plane shear stress*. We get**

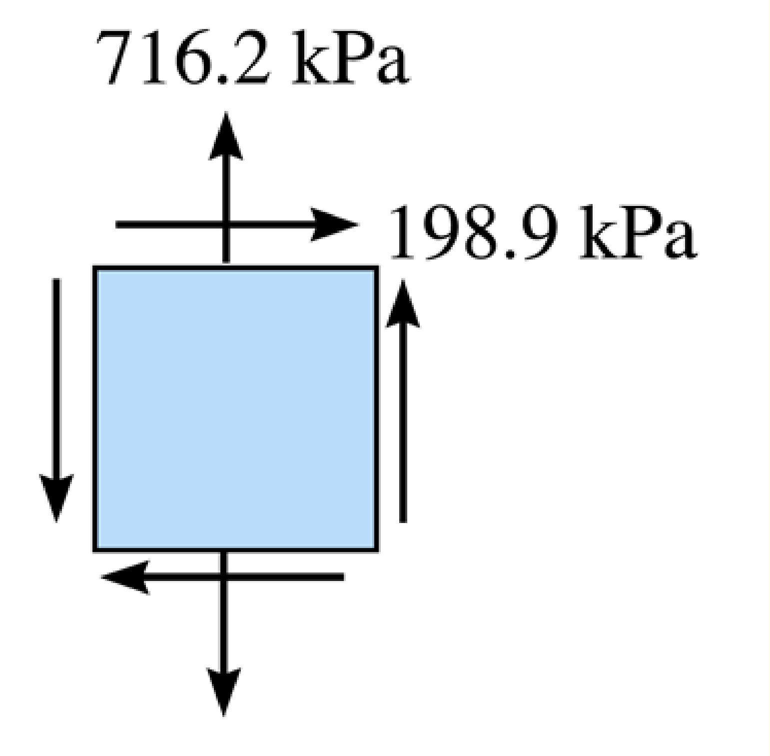
$$\sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}$$




Maximum in-plane shear stresses

# IN-CLASS EXAMPLE 2

The plane-stress state of a point is represented by the element with normal stress and shear stress shown. Represent this state of stress in terms of its principal stresses.



# IN-CLASS EXAMPLE 2, Cont'd

## Solutions

Step 1: Orientation of Element

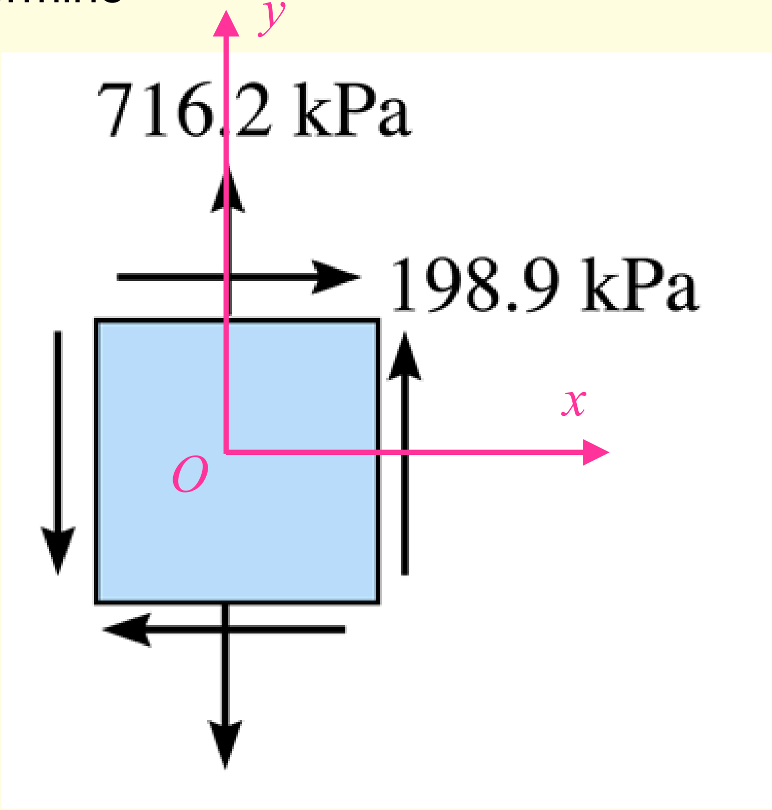
Given $\sigma_x = 0$, $\sigma_y = 716.2\ \text{kPa}$, $\tau_{xy} = 198.9\ \text{kPa}$,

Applying Eq. (9-4) in the textbook p.453 to determine the angle (orientation) of the principal plane

$$2\theta_p = \tan^{-1}\left\{\frac{\tau_{xy}}{(-\sigma_y)/2}\right\}$$

$$2\theta_{p_2} = -29^\circ \qquad \theta_{p_2} = -14.5^\circ$$

$$2\theta_{p_1} = 180^\circ + 2\theta_{p_2} = 151^\circ \qquad \theta_{p_1} = 75.5^\circ$$

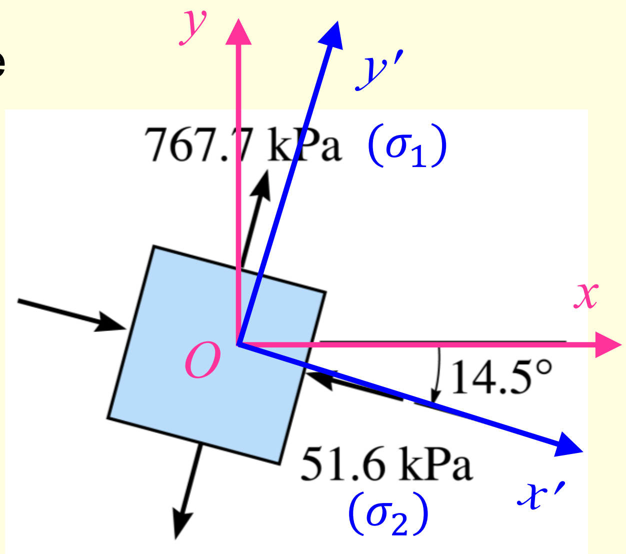


# IN-CLASS EXAMPLE 2, Cont'd

## Solutions

### Step 2: Principal Stress

Applying Eq. (9-5) in the textbook p.454 to determine the principal stresses:




$$\sigma_{1,2} = \frac{\sigma_y}{2} \pm \sqrt{\left(\frac{-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

We get $\sigma_1 = 767.8$ kPa **(Ans.)**

$\sigma_2 = -51.6$ kPa **(Ans.)**

The principal plane on which each normal stress acts on can be determined by applying Eq. (9-1) with, say, $\theta = \theta_{p_2} = -14.5^\circ$. We have

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$
$$= -51.6\ \mathrm{kPa}$$

Hence, $\sigma_2 = -51.6\ \mathrm{kPa}$ acts on the plane defined by $\theta_{p_2} = -14.5^\circ$, whereas $\sigma_1 = 767.8\ \mathrm{kPa}$ acts on the plane defined by $\theta_{p_1} = 75.5^\circ$.

# RELEVANT PRACTICE

- **Assignment 3 on Blackboard, problem #3 (go to Blackboard under the "Assignments" folder, Assignment 3 due: 8:30 AM, Friday, Nov. 5$^{th}$ 2021)**

# OUT-OF-CLASS READING

- **Textbook, pp. 445 – 459**
  - **In particular, Examples 9.2 – 9.6**